MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examinations 2013
Higher 2

CANDIDATE NAME	

CIVICS GROUP	1	2	S					INDEX NUMBER		

H2 BIOLOGY

9648/02

Paper 2 Core Paper

18 September 2013

2 hours

Additional Materials: Answer papers

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid/tape.

Section A
Answer **all** questions.

Section B
Answer any **one** question.

At the end of the examination,
1. Fasten your answer papers to section B securely together.
2. Hand in the following separately:
   - Section A (Part I)
   - Section A (Part II)
   - Section B

The number of marks is given in brackets [ ] at the end of each question or part question.

ANSWER SCHEME

For examiner's Use	
Section A	
1	/ 10
2	/ 12
3	/ 12
4	/ 14
5	/ 9
6	/ 13
7	/ 10
Section B	
8 / 9	/ 20
Total	/100

This paper consists of ___ printed pages.

[Turn over]

Section A (Part I)

Answer **all** the questions in this section.

QUESTION 1

Heavy metal ions are waste products of many industrial processes and at times may leak into the ocean and cause environmental pollution. The toxicity of heavy metals on organisms stems from their effect on metabolically-important enzymes, such as lactate dehydrogenase. Lactate dehydrogenase catalyses the conversion of pyruvate into lactate.

Fig. 1.1 shows the effect of varying concentrations of mercury ions on the rate of reaction catalysed by lactate dehydrogenase isolated from *Oreochromis mossambicus,* a species of fish.

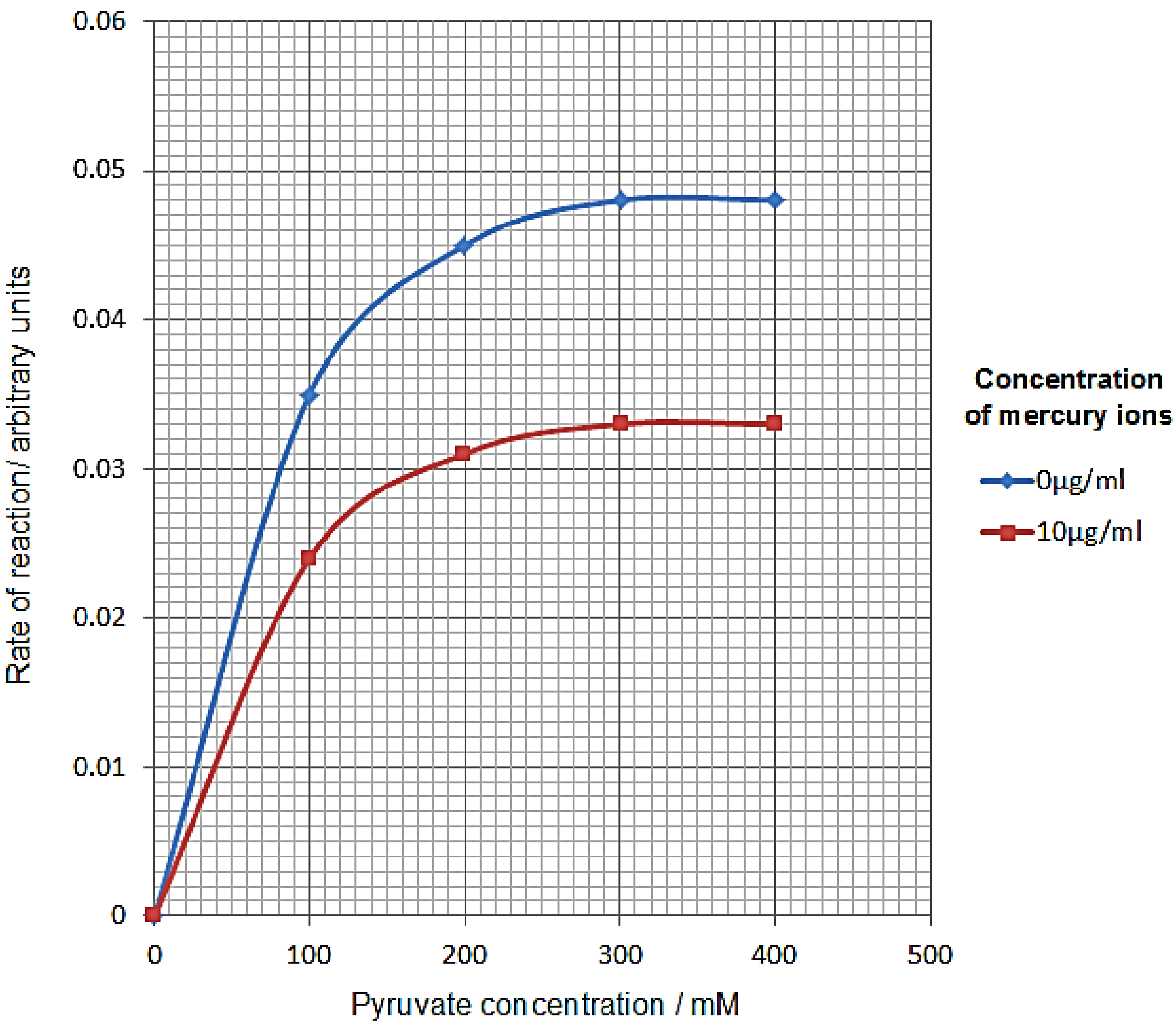

Fig. 1.1

(a) Describe and explain the effect of mercury ions on the rate of reaction catalysed by lactate dehydrogenase. [4]

- In the presence of 10µg/ml of mercury, the **maximum rate (Vmax)** of reaction **decreased** from **0.048 arbitrary units to 0.033 arbitrary units**.
- Mercury ions act as a **non-competitive inhibitor**…
- Has a different structure from the substrate pyruvate and **binds permanently** / **irreversibly** to **another site** away from the active site on the enzyme

  Forms a **strong covalent bond** to an amino acid R group (e.g. -SH group) on the enzyme.
- This causes a **conformational change** on the enzyme which **alters the shape of the active site**.
- Hence the enzyme is rendered non-functional / no longer able to bind to substrate / **decreased rate of formation of enzyme-substrate complex** and hence decreasing the rate of reaction.
- Increasing substrate/pyruvate concentration will not allow the maximum rate of reaction to be achieved.

(b) Explain how mercury ions may affect the survival of the fish when oxygen concentrations are low. [3]

- When oxygen concentrations are low, **aerobic respiration cannot produce sufficient ATP**.
- In addition, the **fish cannot carry out anaerobic respiration** in the presence of mercury ions…
- … as **lactate dehydrogenase** cannot catalyse the conversion of pyruvate to lactate/lactic acid to **regenerate NAD+** in the cytoplasm due to inhibition by mercury ions.
- Hence, **glycolysis** cannot continue to produce **ATP** via **substrate level phosphorylation.**
- Thus **ATP-requiring cellular activities** are **inhibited**, causing the **death** of the fish.

Human lysozyme is an important enzyme which is part of the human body's defence against some bacteria. It is found in abundance in tears, saliva and mucous. The main action of lysozyme is to catalyse the hydrolysis of β-(1,4) glycosidic bonds between N-acetylmuramate (NAM) and N-acetylglucosamine (NAG) sugar residues in the peptidoglycan cell walls of bacteria. With the weakened cell wall, the bacterial cells are more likely to undergo lysis.

Fig. 1.2 shows the structure of lysozyme.

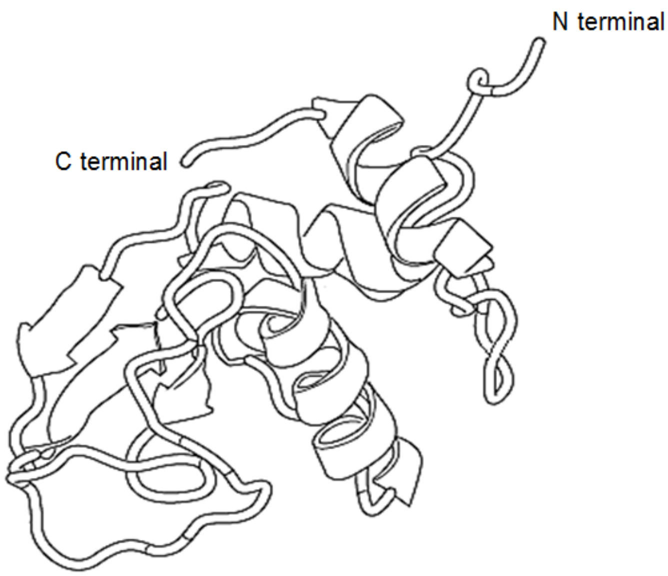

Fig. 1.2

(c) With reference to Fig. 1.2 and with your knowledge of haemoglobin, state two structural differences between lysozyme and haemoglobin. [2]

- The **secondary structure** of lysozyme consists of **α-helices and β-pleated sheets,** whereas the secondary structure of haemoglobin consists of **only α-helices.**
- Lysozyme is a protein with a **tertiary structure** as it consists of only **one polypeptide chain,** whereas haemoglobin is a protein with a **quaternary structure** as it consists of **four polypeptide chains**.
- Lysozyme **does not carry a prosthetic group**, whereas each polypeptide chain in haemoglobin carries a **prosthetic** (non-proteinaceous) **haem group** containing Fe^{2+} which binds a molecule of oxygen each.

(d) Suggest why some bacteria are more resistant to digestion by lysozymes. [1]

- These bacteria cells have a **slime layer** that makes the β-(1,4) glycosidic bonds **less accessible** to the lysozymes.
- These bacteria cells (gram negative bacteria) have a **lower proportion of peptidoglycan** in their cell walls.
- These bacteria (gram negative bacteria) have an additional **outer membrane** that makes the β-(1,4) glycosidic bonds **less accessible** to the lysozymes.
- AVP (e.g. different set of monomers/bond making up the cell wall)

[Total: 10]

QUESTION 2
Fig. 2.1 illustrates the Central Dogma, which describes the flow of genetic information from DNA to RNA to proteins.

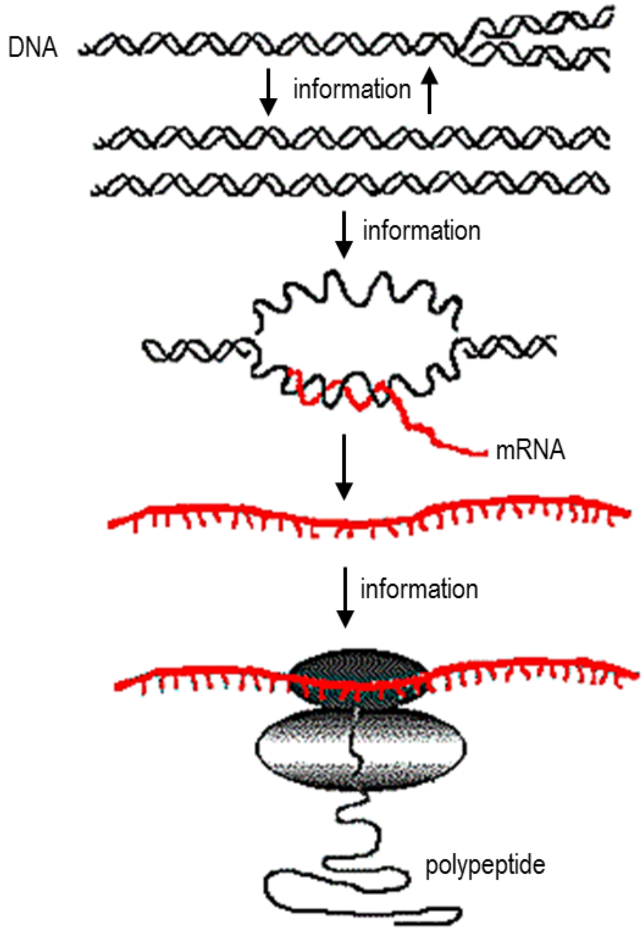

Fig. 2.1

(a) Explain, with an example, how the Central Dogma may not always hold true. [2]

- There can be reverse **flow of genetic information** from **RNA to DNA**
- E.g. in **HIV** virus, **reverse transcriptase** converts viral RNA to DNA / **reverse transcription** occurs in HIV

- **RNA** instead of DNA can store **genetic information**
- E.g. the **influenza** virus has several segments of linear **(–) single stranded RNA** genome.

- **RNA** can undergo **replication**.
- E.g. **influenza** virus contains **RNA-dependent RNA polymerase** **synthesizes (+)RNA using (–)RNA as the template** via complementary base pairing.

(b) Describe how the information carried on mRNA allows the synthesis of a complete polypeptide chain. [3]

- **5' UTR** allows **recognition by small ribosomal subunit** / **translation factors**
- **Start codon** signals the start of translation
- **Three bases** on the mRNA correspond to a **codon**
- **Each codon** on mRNA codes for a specific **amino acid**
- The **stop codon** (UAG, UAA, UGA) **terminates translation** by allowing the binding of a release factor that adds a water molecule to the polypeptide chain

A ribonucleoprotein is a protein that is associated with RNA. The ribosome is an example of a ribonucleoprotein. Prokaryotes have 70S ribosomes while eukaryotes have 80S ribosomes, each consisting of a small and large subunit. In prokaryotes, the small subunit is made up of a 16S rRNA associated with 21 proteins. Fig. 2.2 shows the structure of a 16S rRNA.

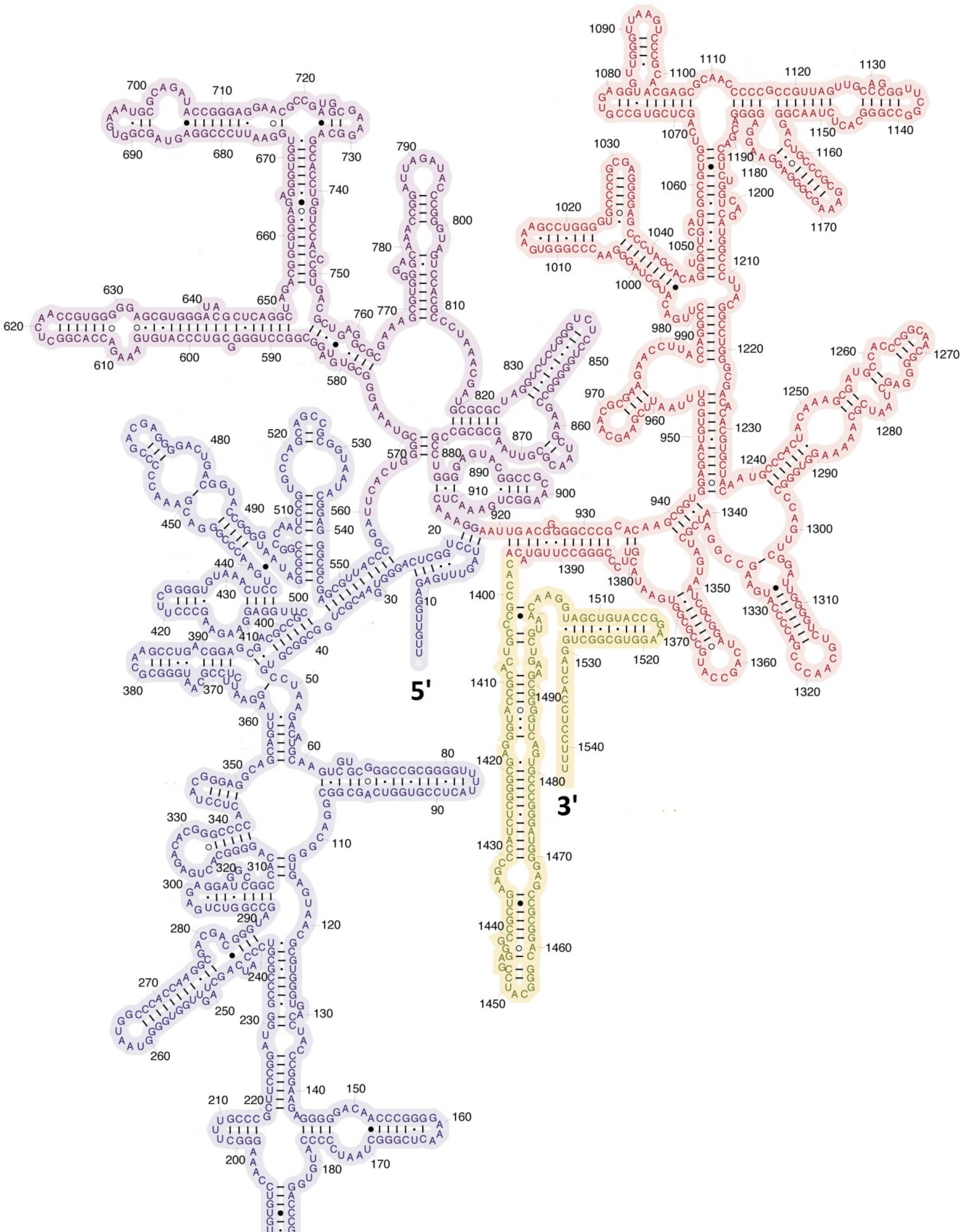

Fig. 2.2

(c) 16S rRNA and tRNA are both made up of four types of ribonucleotides linked by phosphodiester bonds.

State two other similarities between the structures of the 16S rRNA and tRNA. [2]

- Both rRNA and tRNA are **single stranded** molecules.
- Both have **double-stranded regions** / can **form loops**
- Both have intra-molecular **hydrogen bonds** between **complementary bases** / between A&U and C&G.

(d) Describe how eukaryotic ribosomes are formed. [4]

- **rRNA** is produced through **transcription** of the **rRNA genes** in the **nucleolus**
- **Genes of ribosomal proteins** are transcribed in **nucleus** to form **mRNA**
- which is **translated** in **cytoplasm** to form **ribosomal proteins**
- **Ribosomal protein** is **transported from the cytoplasm into nucleus** via nuclear pore
- Partial **assembly of rRNA and ribosomal proteins** into large and small subunits occurs in **nucleolus**
- The **large and small subunits** are **transported to the cytoplasm**

Unlike eukaryotes, prokaryotes do not have endoplasmic reticulum or vesicles.

(e) Suggest how prokaryotes are still able to secrete proteins. [1]

- Prokaryotes have **protein channels** in the cell surface membrane that allows secretion of proteins

[Total: 12]

QUESTION 3
The Human Immunodeficiency Virus (HIV) weakens the immune system, causing the patient to succumb to secondary infections, leading to the development of AIDS. The first recognized case of AIDS was reported in 1981 in Los Angeles, U.S.A.

Fig. 3.1 shows the binding of a HIV particle to a T cell. The viral receptor gp120 trimer binds to both CD4 and a co-receptor called CCR5 on the T cell surface.

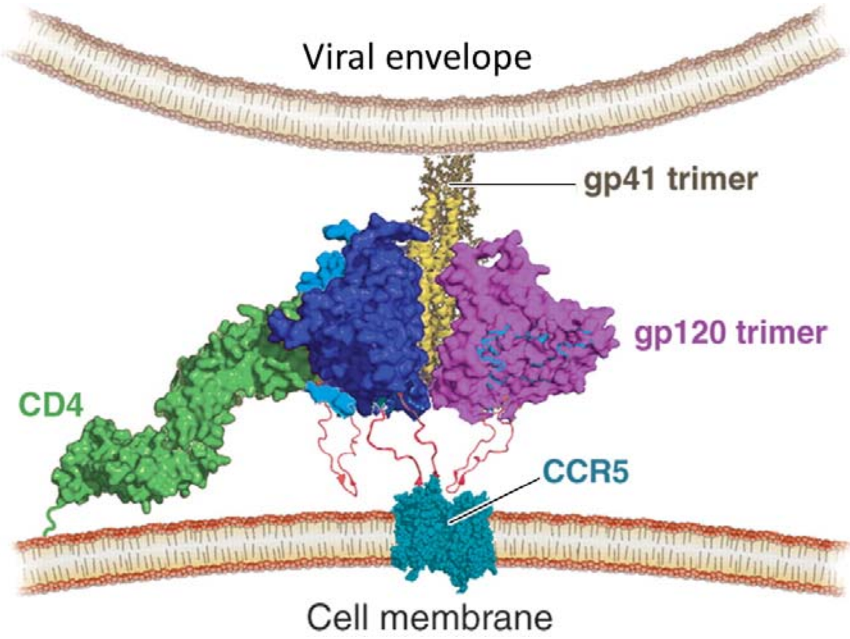

Fig. 3.1

(a) Explain why the composition of the viral envelope is similar to the composition of the T cell membrane. [1]

- Viral membrane is **derived from the host cell** when the virus **exits the host cell** via **budding**

(b) Individuals who carry a mutation known as CCR5–Δ32 in both copies of the CCR5 gene are resistant to certain strains of HIV. The CCR5–Δ32 allele is a result of a deletion of a particular sequence of 32 base-pairs within the exon of CCR5 gene.

Explain how an individual who is homozygous for the CCR5–Δ32 allele is resistant to HIV. [4]

- **Loss-of-function mutation**
- The deletion causes a **conformational change / mis-folding** in the CCR5 co-receptor / gp120 binding domain of CCR5
- The **shape of the mutant CCR5 co-receptor no longer complementary** to the **shape of gp120** hence cannot bind
- Individual is homozygous – hence no normal CCR5 co-receptor is produced at all.
- **Entry** of HIV is **inhibited**
- **Loss-of-function mutation**
- The deletion causes **mis-folding** of the CCR5 co-receptor, leading to its **degradation**.
- **Absence** of **CCR5 co-receptor** on the cell surface
- Individual is homozygous – hence no normal CCR5 co-receptor is produced at all.
- **Entry** of HIV is **inhibited**

(c) It was observed that this strain of HIV that uses CCR5 as the co-receptor quickly evolves into another strain which bind to another co-receptor known as CXCR4.

Suggest and explain how HIV can evolve to use CXCR4 as the co-receptor. [2]

- **HIV reverse transcriptase** is **error-prone** / has **no proofreading function**
- Results in **mutation** to **gp120 gene**
- The conformation of the mutated gp120 protein is complementary to and binds CCXR4

(d) The allele frequency of CCR5-Δ32 is unusually high in the western European countries, with up to 20% of ethnic western Europeans carrying this allele, which is rare or absent in all other ethnic groups throughout the world.

Suggest a reason for this phenomenon. [1]

- **Founders' effect** – a small group of individuals from a larger population, some of whom carry the CCR5-Δ32 allele, colonized/established a new population in the western European countries.

(e) A technique called RNA interference has been shown to slow down the replication of HIV *in vitro*. Double-stranded RNA molecules, 21 to 23 nucleotides long, were added to a culture of T cells infected with HIV. The sequence of this small interfering RNA (siRNA) matches part of the HIV protease gene. Once inside the T cell, the two strands of the siRNA separate into single strands. One of the strands is identical in sequence to the mRNA of the HIV protease.

Explain how this siRNA is able to interfere with HIV replication. [4]

- The other strand of the siRNA is **complementary** to and **binds** to the **mRNA** of HIV protease
- Interferes with **translation** of the HIV protease mRNA into the enzyme protease, hence new HIV protease is not produced.
- HIV protease not packaged into new HIV viruses
- During subsequent infection, HIV **polyprotein** is **not cleaved** into the functional HIV polypeptides
- Interferes with **assembly** of HIV particles.
- One strand of the siRNA is **complementary** to and **binds** to the **(+) RNA** of HIV protease.
- Interferes with the **reverse transcription** of (+)RNA to form single-stranded DNA
- No double-stranded DNA is formed and hence **no integration** of dsDNA of HIV into host genome
- No new (+)RNA genome / HIV mRNA will be formed, hence interfering with assembly.

[Total: 12]

QUESTION 4

Gene expression in eukaryotic cells can be controlled in many ways, one of which is the regulation of mRNA production. Fig. 4.1 shows how transcription can be controlled.

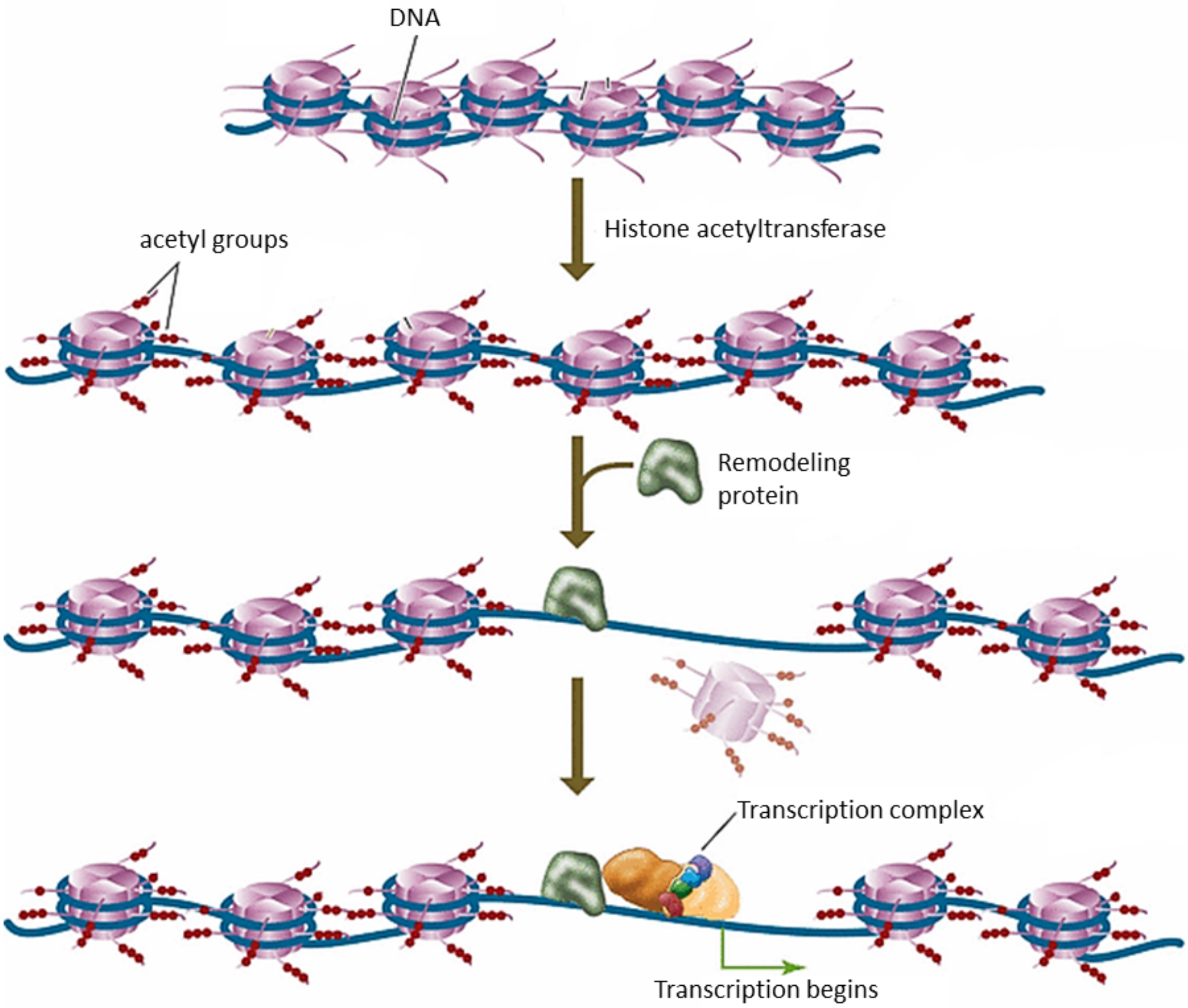

Fig. 4.1

(a) Using the information in Fig. 4.1 and your knowledge of the control of eukaryotic gene expression, explain how transcription is initiated. [4]

- **Histone acetyltransferase** **neutralizes** positive charge on lysine residues on the **histone tail** by **adding acetyl groups**
- **Loosen the attraction** between the **negatively-charged DNA** and octameric histone
- **Remodeling protein** **dissociates/disaggregates**/ octameric histone from the DNA
- Allows the **assembly of transcription complex** on the promoter / **binding** of transcription factors and RNA polymerase to the promoter to initiate transcription

Ubiquitin-dependent protein degradation in the proteasome is one way by which the amount of cytosolic proteins in a cell can be controlled at the post-translational level. An amino acid sequence of proline, glutamic acid, serine and threonine (PEST) in a protein are often recognized by enzymes and tagged with ubiquitin proteins. Globular proteins in their correct conformation are not tagged for degradation.

(b) Using the above information, suggest why denatured cytosolic proteins can be tagged with ubiquitin. [1]

- Denaturation **exposes** the hidden PEST sequence within the hydrophobic core of the protein to be tagged with ubiquitin by enzymes.

(c) Transmembrane proteins are usually not degraded by proteasome.

Suggest how transmembrane proteins on the cell surface are degraded. [2]

- **Endocytosis** of membrane proteins resulting in formation of **vesicles/endosome**
- **Fusion** of vesicles with **lysosomes** degrade the membrane proteins

Ovalbumin gene from chicken was isolated and made single-stranded, which was then mixed together with its mature mRNA. The results were observed under an electron microscope. Fig. 4.2 shows the electron micrograph and its corresponding diagrammatic representation, showing the binding of the mRNA to certain regions of the DNA.

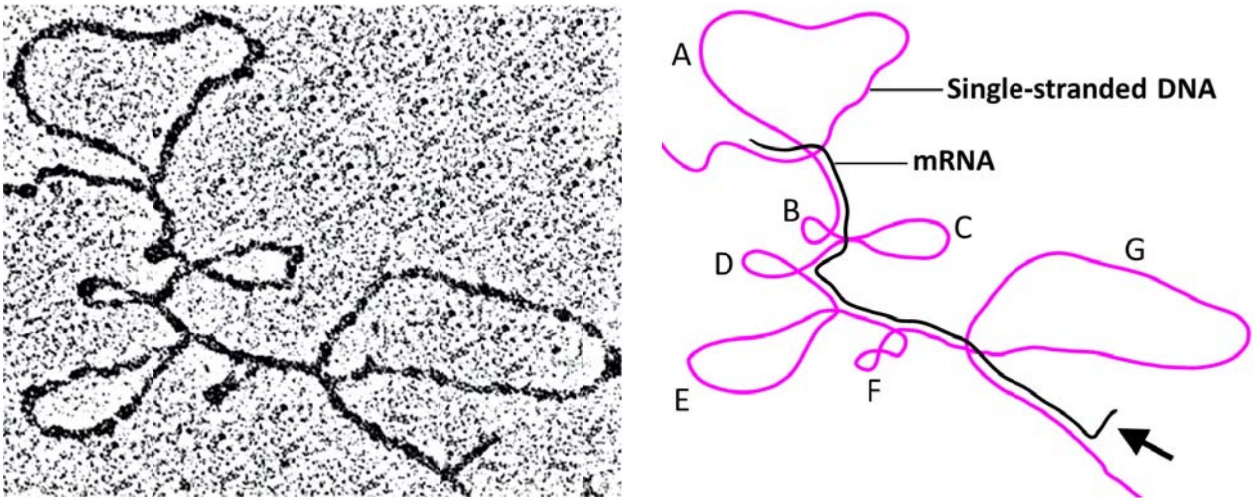

Fig. 4.2

(d) Explain why loops (A – G) were observed. [3]

- The single-stranded DNA is the **template strand** from which the mRNA is transcribed.
- The **loops are the introns** on the DNA that are not part of the mature mRNA sequence.
- This is because in mature (cytosolic) mRNA, the **introns are excised** and **exons are spliced** together during post-transcriptional modification.
- The mRNA **can only bind to** / **base pair with the exons** on the template DNA.

(e) Explain why the region of the mRNA, indicated by the arrow in Fig. 4.2, remains unbound to DNA. [2]

- It is the **3' end** of the mRNA where the **poly(A) tail** was added during post-transcriptional modification.

- There is no corresponding stretch of thymine on the template DNA where the poly(A) tail can form complementary base pair with / not complementary to the DNA template

Repeating nucleotide sequences are common in the genome of eukaryotes. These repeating sequences have been commonly referred to as ‘junk DNA’.

(f) Suggest why the term ‘junk DNA’ is misleading. [2]

- ‘Junk’ implies no function / some repeating sequence do have a function
- Centromere – site of kinetochore assembly for attachment of microtubule / hold two sister chromatids together
- Telomere – ref. any one function of telomere
- Regions to break and rejoin during crossing over / regions for chiasmata formation
- Buffer against mutation on coding regions
- AVP

[Total: 14]

QUESTION 5

Hereditary hypophosphatemic rickets is a genetic disorder that results in low level of phosphate in the blood (hypophosphatemia).

Fig. 5.1 shows the inheritance of this disease over four generations in an extended family.

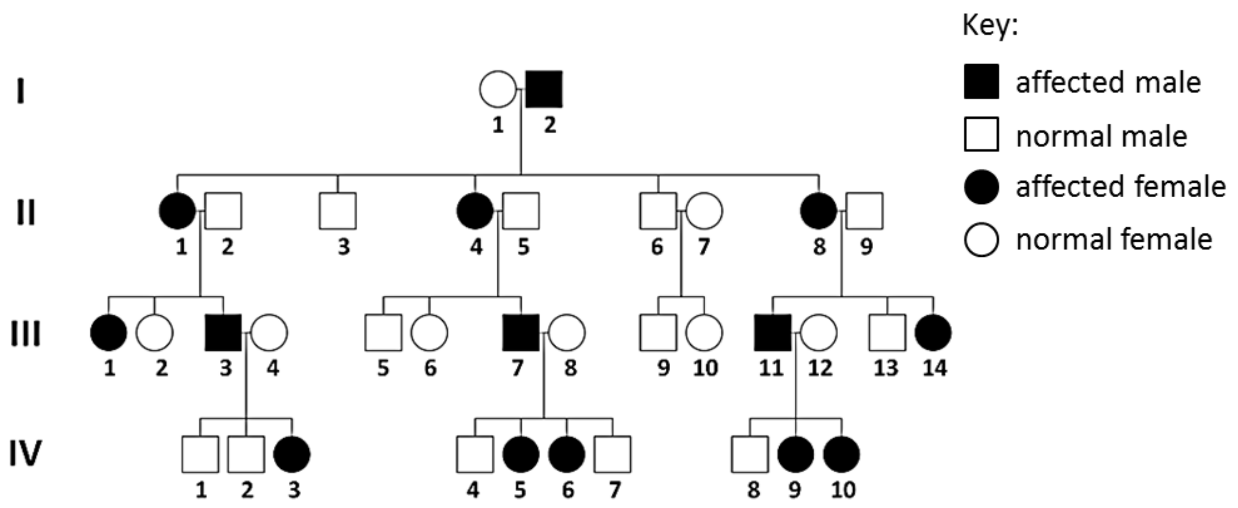

Fig. 5.1

(a) Explain the mode of inheritance of this disease. [2]

- **X-linked dominant** / **sex-linked dominant** inheritance
- An affected male pasess the disease allele on the X chromosome to **all his daughters** but **not his son** **+ citing any one example**
    - o I-2 passed the dominant allele to daughters II-1, II-4, II-8 but not to the sons
    - o III-7 passed the dominant allele to daughters IV-5 and IV-6 but not to the sons
    - o III-3 passed the dominant allele to daughters IV-3 but not to the sons
    - o III-11 passed the dominant allele to daughters IV-9 and IV-10 but not to the sons

(b) Using the symbols **D/d** to represent the alleles, construct a genetic diagram to show how individual **III-7** and **III-8** can pass on the disease to their children. [4]

Parental phenotype	Affected male		Unaffected female		
Parental genotype	X^DY		X^dX^d		[1]
Gametes	X^D	Y	X^d	X^d	[1]
F1 genotype	X^DX^d	X^DX^d	X^dY	X^dY	
F1 phenotype	Affected female	Affected female	Unaffected male	Unaffected male	[1]
F1 phenotypic ratio	1 affected female : 1 unaffected male				[1]

Fur colour of Labrador dogs is controlled by two genes, **B/b** and **E/e**. The genes are on different pairs of chromosomes.

Allele **B** produces black fur while allele **b** produces brown fur. However, the colour is expressed only in the presence of allele **E**. In the absence of allele **E**, the dogs are beige in colour, hence the name Golden Labrador.

Crossing two heterozygotes typically gives the following phenotypic ratio:

9 black : 3 brown : 4 golden

(c) Explain how the phenotypic ratio will differ if genes **B/b** and **E/e** are located on the same homologous pair of chromosomes, as shown in Fig. 5.2. [3]

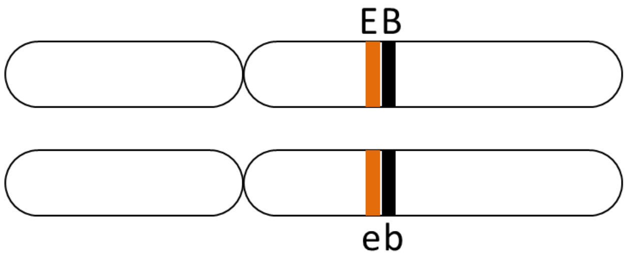

Fig. 5.2

- 3 black : 1 golden
- **Tightly** linked, hence **no crossing over** / **chiasma not formed** between gene **B/b** and **E/e**.
- Inherited as one unit
- No recombinant gametes

- No fixed ratio
- Large number of black and golden, small number of brown dogs
- **Closely linked**, hence **crossing over** between gene **B/b** and **E/e** occurs at low frequency.
- Small number of recombinant gametes

[Total: 9]

Candidate Name | Civics Group | Index Number

Meridian Junior College
JC2 Preliminary Examinations 2013
H2 Biology (9648/02)

Question 6	Question 7
/ 13	/ 10

Section A (Part II)
Answer **all** the questions in this section.

QUESTION 6

(a) Describe the role of the following ions in impulse transmission along a nerve cell.

(i) Na^{+} [2]

- Influx of/ movement of Na+ ions **into the axoplasm** via facilitated diffusion through voltage-gated Na+ channels and causes **depolarization**
- Trigger **action potential** and brings membrane potential to **+40 mv**

(ii) K^{+} [2]

- Efflux/movement of K+ ions **out of axoplasm** and causes **repolarization**
- Restores the **resting potential** to **-70 mV**

(b) Describe the role of Ca^{2+} in the passage of impulses across a synapse. [3]

- **Ca^{2+} influx** into presynaptic neuron via facilitated diffusion through voltage-gated Ca^{2+} channels
- The increase in Ca^{2+} causes **vesicles that contain neurotransmitters** to move to and **fuse** with **presynaptic membrane**
- causes the **neurotransmitter to be released into the synaptic cleft** by **exocytosis**

(c) Explain why the concentration of Ca^{2+} in the cytosol of the synaptic knob remains low despite frequent opening of channels. [2]

- There are **carrier protein** / **protein pump** on the pre-synaptic membrane…
- …that **actively transport** / **pump Ca^{2+} out of the cytosol** of synaptic knob at the expense of **ATP**.

A water-soluble toxin, muscarine, derived from the mushroom, causes deceleration of the heart rate, resulting in death of individuals. Fig. 6.1 shows the signalling pathway of muscarine on postsynaptic membrane.

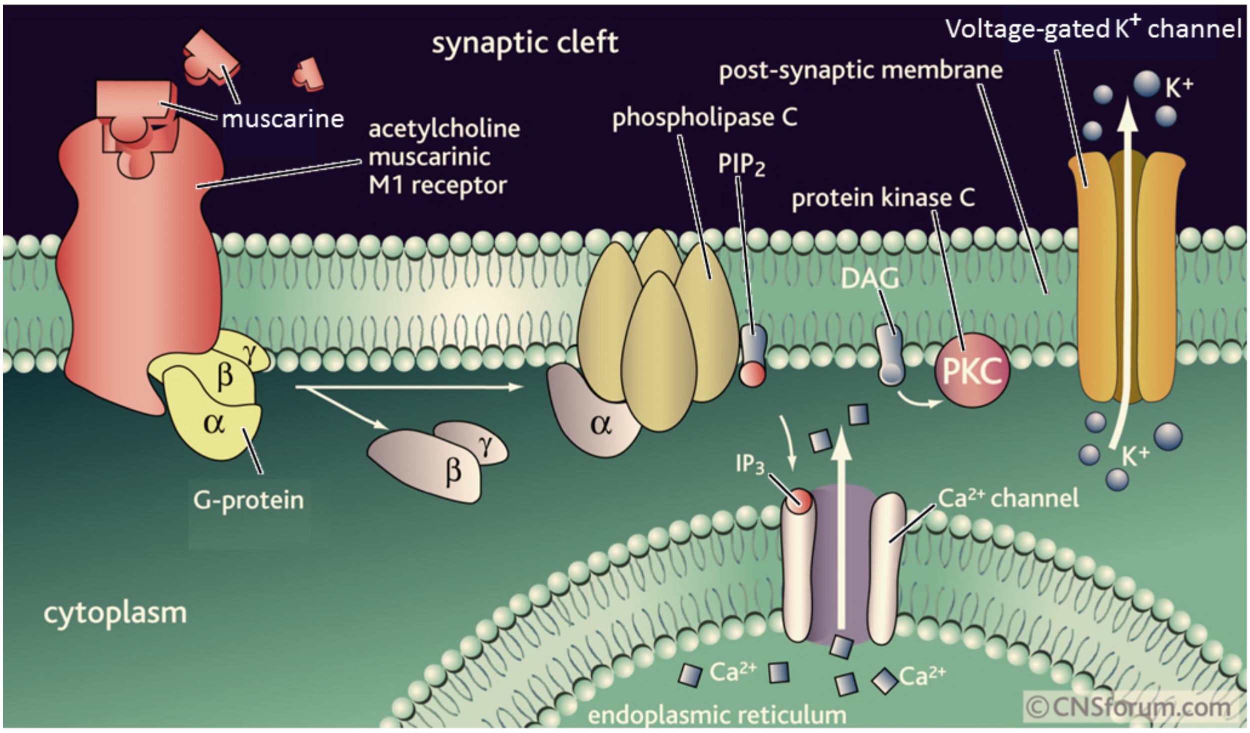

Fig. 6.1

(d) Describe how muscarine results in inhibition of nerve transmission in heart muscle cells. [4]

1. **Muscarine binds to acetylcholine muscarinic receptor** to cause the **activation of G-protein by exchanging GDP for GTP**

2. The activated G protein **activates phospholipase C that convert PIP_2** to second messengers, **IP_3 and DAG**.

3. IP3 binds to the **IP_3/ligand-gated Ca^{2+} channel** and cause the **opening of Ca^{2+} channels on the ER**, resulting in **the release of Ca^{2+} ions into the cytosol**.

4. This **increase the membrane potential** causing the **opening of voltage-gated K^+ channel**, and hence the **efflux of K^+ ions into the synaptic cleft**

5. This results in **hyperpolarisation** / **inhibitory postsynaptic potential (IPSP)** in heart muscle cells.

[Total: 13]

QUESTION 7
Fig. 7.1 shows the distribution of malaria and the allele frequency of the sickle cell allele.

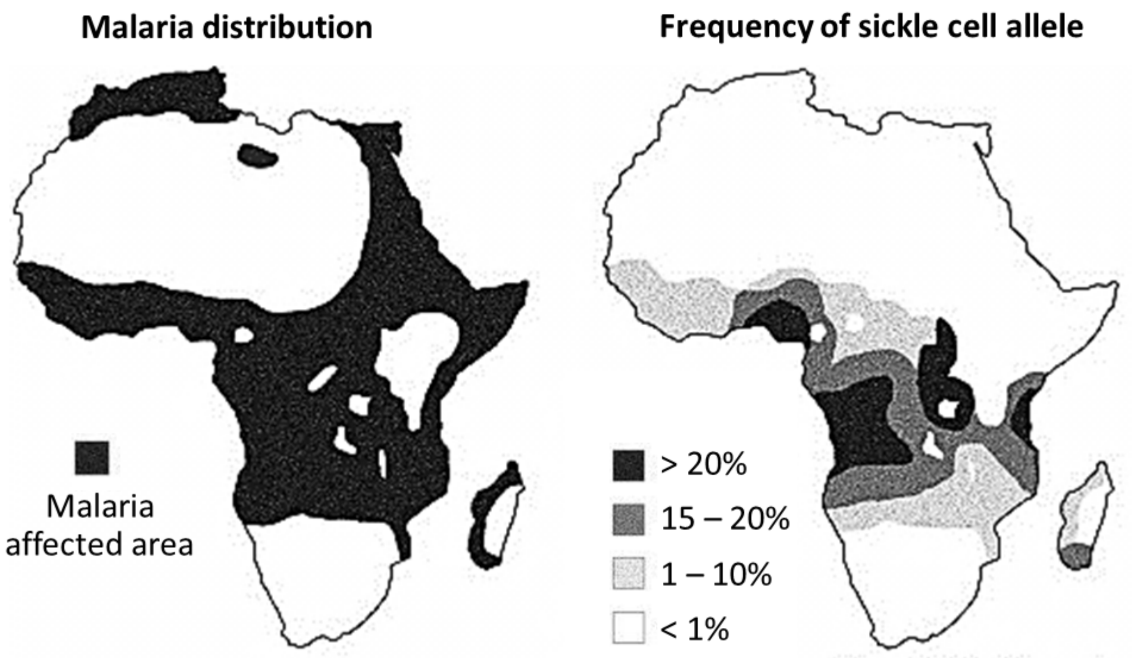

Fig. 7.1

(a) Describe and explain the correlation between the occurrence of malaria and the frequency of the sickle cell allele. [4]

- **Frequency of sickle cell allele** is **higher** in **malaria affected areas**
- Heterozygotes do not exhibit sickle cell anaemia / phenotypically normal
- Heterozygous / sickle cell traits individuals are at **selective advantage** in malaria affected areas / more resistant to malaria
- Heterozygous individuals **survive to reproduce** and **pass on the sickle cell allele** to **viable and fertile offspring**…
- …hence **increasing the HbS allele frequencies in the population** within these malaria areas over time.

The evolution of haemoglobin molecules has been studied extensively by comparing the amino acid sequences in both myoglobin and haemoglobin. Myoglobin is used for oxygen storage while haemoglobin is used for oxygen transport. Ancient prehistoric animals had a single chain of simple globin for oxygen storage and transport.

About 500 million years ago, a gene duplication event occurred and one copy became the present day myoglobin and the other evolved into an oxygen-transport protein that gave rise to the present day haemoglobin (Fig. 7.2).

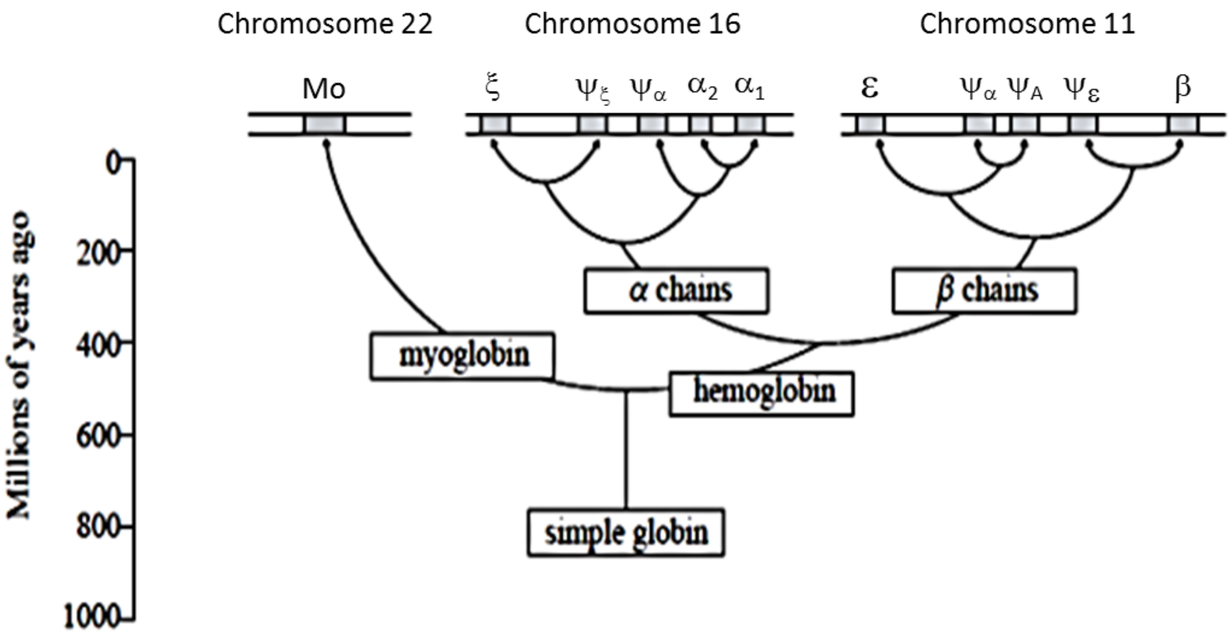

Fig. 7.2

(b) State how many years ago the haemoglobin molecule diverged into α chains and β chains. [1]

- 400 million years ago

(c) Suggest why changes observed in the sequence of amino acids may lead to an under-estimation of the actual number of mutations. [2]

- A mutation may result in the **same amino acid** being produced due to **genetic code** being **degenerate**
- Some mutations may have occurred in the **non-coding DNA region** / **introns**

(d) Explain how variations in the haemoglobin gene can be used as a molecular clock. [3]

- The **mutation** in the gene coding for haemoglobin occurred at a **constant rate** through time
- The number of nucleotide differences can be used as a gauge to estimate when two species diverged from a common ancestor / **OWTTE**
- The **lesser the number of nucleotide difference**, the **more closely related** the two species are [**A:** reverse argument]

[Total: 10]

Section B
Answer **ONE** question

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labeled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be in set out in questions **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 8

(a) Describe the importance of ATP in cellular processes. [8]

1. **Hydrolysis** of ATP **releases energy**

2. **Movement of chromosomes** to the metaphase plate during nuclear division.

3. **Movement of transport vesicles** from the rER to Golgi apparatus / from the Golgi apparatus to the cell surface membrane

4. **Proton pumps** on the lysosomal membrane, which pumps H+ from the cytosol into the lysosome, to **maintain its acidic pH**

5. **Sodium and potassium ion pump** on the cell surface membrane of axons **maintains the resting potential** of neurons. Energy of one ATP molecule is used to actively pump three sodium ions out to the tissue fluid and two potassium ions in to the axoplasm.

6. **Calcium pumps** for the active transport of calcium ions out of synaptic knob. This **maintains a low concentration of calcium ions in the synaptic knob**, thus regulating synaptic transmission.

7. Active **reabsorption of choline** back to the synaptic knob for the **reformation of acetylcholine** neurotransmitter. This allows the cell to save cellular resources.

8. In **endocytosis** / **phagocytosis** / **pinocytosis** / **receptor-mediated endocytosis**, a cell brings in macromolecules/particulate matter by forming new vesicles from the plasma membrane by forming **pseudopodia** / **invagination of membrane**

9. In light-independent reactions/Calvin cycle of photosynthesis, ATP is required for the **reduction of glycerate-3-phosphate (GP) to glyceraldehydes-3-phosphate (GALP)**,

10. and for the **regeneration of ribulose bisphosphate** (RuBP).

11. In the initial stages of glycolysis in respiration, ATP is required to donate phosphate groups during the phosphorylation of **glucose to glucose-6-phosphate**

12. **fructose-6-phosphate to fructose-1,6-bisphosphate**.

13. Form aminoacyl-AMP-enzyme complex, catalysed by the enzyme aminoacyl-tRNA synthetase. The complex then attaches to a specific tRNA to form aminoacyl-tRNA complex/activated amino acid.

14. In cell signalling, ATP donates a phosphate group to a target protein, catalysed by protein kinases.

15. This causes the target protein to undergo a conformational change and **convert from the inactive to active form**, which leads to signal transduction.

16. ATP is also used as a **substrate** for the **formation of cAMP** by **adenylyl cyclase**. cAMP acts as second messenger that leads to signal transduction.

(b) Distinguish between a specialized cell and a cancerous cell. [5]

	Property	Specialized cell	Cancerous cells
1	Differentiated?	Differentiated with specialized function.	Remains undifferentiated with no specialized function
2	Contact inhibition	Present. Cells do not divide further when in contact with other cells.	Absent. Cells continue to divide even when in contact with neighbouring cells.
3	No. of cell division	Cells divide for a limited number of times (Hayflick limit)	Cells divide infinitely and uncontrollably.
4	Apoptosis	Undergo apoptosis	Does not undergo apoptosis
5	Angiogenesis (growth of blood vessels towards itself)	Occurs only when new tissue is needed to repair damaged tissue.	Occurs even when growth is not necessary.
6	Proto-oncogene / oncogene	Proto-oncogenes code for proteins that stimulate normal cell growth and division.	Oncogenes code for hyperactive/excessive proteins that cause excessive cell division.
7	Tumour suppressor gene	Functional tumour suppressor gene code for proteins that prevent uncontrolled cell division.	Mutated tumour suppressor gene code does not code for a protein / code for non-functional / less-functional protein, hence unable to prevent uncontrolled cell division.
8	Telomerase gene expression	Telomerase enzyme absent / gene not expressed	Telomerase enzyme present / gene expressed
9	AVP	AVP	AVP

(c) Discuss, with a named example, how mutation to centromeres can result in genetic disorders. [7]

1. Mutation to the centromere **changes the shape/3D conformation** of the centromeres due to changes in nucleotide sequence.

2. This **prevents** the **binding of kinetochore protein complex to the centromere**…

3. …as the **kinetochore protein binding site** is **no longer complementary in shape to the centromeres**.

4. This disallows the **attachment of spindle fibres** / **microtubules**.

5. Nondisjunction occurs as the **separation of homologous chromosomes** during **anaphase I** does not occur.

6. Nondisjunction occurs as the **separation of sister chromatids** during **anaphase II** does not occur.

7. Nuclear membrane reforms around the abnormal number of chromosomes.

8. This results in the formation of **abnormal gametes** / **(n+1) or (n-1) gametes** during meiosis.

9. When fusion of a **normal haploid gamete (n)** from one parent and an **abnormal gamete with (n+1) or (n-1) chromosomes** from another parent occurs, this results in a genetic disorder.

10. For example, **Down's Syndrome** is a condition where there is presence of an **extra chromosome 21** / **any valid examples**

QUESTION 9

(a) Describe and explain how the structures of chloroplasts and mitochondria are similarly adapted for their functions. [8]

	Structural similarity	Function
1	The **presence** of **electron carriers** on the **thylakoid membrane** and **mitochondrial inner membrane**	Passage of electrons via the electron carriers **release energy** to create proton gradient for the synthesis of ATP
2	The **presence** of **proton pump** on the thylakoid membrane and mitochondrial inner membrane	**Pump protons** from the stroma to thylakoid space in chloroplast and matrix to inter-membrane in mitochondria to create a **proton gradient**
3	The **inner membrane of mitochondria** and the **thylakoid membrane** is **extensively folded**	To **increase surface area** for attachment of e.g. electron carriers / photosynthetic pigments
4	Both contain **ATP synthase** on the thylakoid membrane and mitochondrial inner membrane	To allow the **facilitated diffusion of protons** to generate ATP To allow **phosphorylation of ADP to ATP**
5	Both contain **circular DNA**	To contain **genes** that code for proteins/enzymes involved in photosynthesis and respiration
6	Both contains **70S ribosomes**	For **translation** of mRNA into proteins/enzymes involved in photosynthesis and respiration
7	Both are **double-membrane** bound organelles	For **compartmentalization** and localization of enzymes for Krebs Cycle and link reaction in mitochondria matrix and Calvin Cycle in chloroplast stroma

(b) Discuss the factors that contribute to speciation. [7]

1. Speciation is the **formation of new species** / **divergence of existing species into new species**

2. **Spontaneous mutations** give rise to **variation** within the population…

3. …which allow **favourable characteristics to be selected for** and **unfavourable characteristics to be selected against**.

4. Speciation is partly due to **disruption to gene flow** in a population of a particular species.

Reduced gene flow as a result of:

5. **Behavioural isolation**, where in a population in a particular area, some members of the population exhibit variation in behaviour and **tended to mate preferentially**.

6. **Geographical isolation**, where in a **physical barrier** separates members of a population into 2 sub populations, preventing interbreeding.

7. **Physiological isolation**, various mechanical, physical, and biochemical differences between members of a population become barriers to interbreeding between populations, thus maintaining separate, distinct species

8. **Any other isolation, e.g. gametic isolation**

9. **Genetic drift** / **Founder's effect** / **Bottleneck effect** **changes the allele frequencies** within the population by chance.

10. **Accumulation of mutations / differences** over many generations

11. Over many successive generations, each subpopulation became 2 **genetically distinct species** which were **unable to interbreed** and produce fertile viable offspring.

(c) Explain the concept of negative feedback in controlling insulin levels in the body. [5]

1. Negative feedback is a mechanism where the **set point** of a parameter is **restored** by mechanisms that eliminate the disturbance / by moving the parameter in the direction opposite of the disturbance.

2. **α and β cells** of the islet of Langerhans in the pancreas **detect** an **increase** in **blood glucose level** above **set point** of 90mg/100ml.

3. **β cells** of the islet of Langerhans **secrete insulin** and α cells of the islet of Langerhans stop secretion of glucagon.

4. This **decreases** the **blood glucose level** back to the **set point.**

5. **β cells** of the islet of Langerhans **stop secretion of insulin…**

6. … and **circulating insulin** will be **removed from the blood** / destroyed in the liver.

7. The **level of insulin** is **reduced**.

[Total: 20]